

Introduction to WAN Technologies – Exam Notes

What is a WAN?

- Wide Area Network
- Covers large geographical areas
- Uses common carriers (telephone companies)
- Operates mainly at Physical, Data Link, Network layers

Point-to-Point Links

- Dedicated leased line
- Fixed connection between two sites
- High reliability, high cost

Circuit Switching

- Temporary connection
- Call setup → data transfer → termination
- Example: ISDN

Packet Switching

- Shared carrier network
- Data sent as packets
- Cost effective
- Examples: Frame Relay, ATM, X.25

Virtual Circuits

- Logical path in shared WAN
- SVC: On-demand, temporary
- PVC: Permanent, always active

WAN Dialup Services

- DDR: Dial only when needed
- Dial Backup: Backup for main link

WAN Devices

- WAN Switch: Carrier network switching
- Access Server: Dial-in/out concentration
- Modem: Digital ↔ Analog conversion
- CSU/DSU: Router to digital line
- ISDN Terminal Adapter: Digital only